

A self-serve canvas for simple AI automations agent-flow-studio

Situation

Meridian Services Group (illustrative mid-size B2B services company) has an ops team that keeps getting the same shape of request — route these tickets, triage these emails, escalate anything urgent. Each is a small automation, and every one is a ticket in the engineering backlog, because business users can't assemble one themselves.

The problem, in numbers

40

small-automation requests
/ year to engineering

€67k

engineering time / year
on small flows (estimate)

4-6 wk

typical queue wait
before work starts

backlog

simple wins hidden
behind product work

Solution

A visual agent-workflow builder that runs entirely in the browser — no backend, no build step, no install, no API key. A business user drags five node types, wires them, hits Run, and watches a live trace. The same engine.js runs in the browser and under the tests.

Trigger

LLM Agent

Condition

Tool

Output

5 node types, a real topological-sort executor, and 2 example flows that run end-to-end in CI.

Impact / ROI

Illustrative model: move the simple flows off the engineering queue to same-day self-serve. Keep engineering for the genuinely complex ~30%.

28/yr

flows a business user
self-serves (~70%)

672 h

engineering hours
freed / year

€47k

engineering cost
freed / year (est.)

same-day

lead time, down from
4-6 weeks

Recommendation

Use the canvas as a self-serve intake for simple flows and keep engineering for the complex 30%. This repo is the working proof-of-concept of the pattern; a production rollout would put a real model behind the agent node and add auth and retries.

How the numbers were derived

Assumptions (estimates)

Small-automation requests per year	40
Engineering effort each (build + review + deploy)	24 h
Fully-loaded engineer cost	€70 / hour
Share simple enough to self-serve	70%
Typical queue wait before work starts	4-6 weeks

The arithmetic

Eng. hours/year = $40 \times 24 \text{ h} = 960 \text{ h}$

Eng. cost/year = $960 \text{ h} \times €70 = €67,200$

Self-served/year = $40 \times 70\% = 28 \text{ flows}$

Hours freed/year = $28 \times 24 \text{ h} = 672 \text{ h}$

Cost freed/year = $672 \text{ h} \times €70 = €47,040$

Payback = no infra / licensing cost -> first self-served flow

Real repo facts (not estimated)

What the ROI rests on — the working proof-of-concept:

- 16 passing engine tests (node --test), pure Node built-ins, no dependencies
- 5 node types on a browser canvas: Trigger, LLM Agent, Condition, Tool, Output
- a real topological-sort executor (Kahn's algorithm) with cycle detection
- 2 example flows (RFQ triage, support-ticket router) that run end-to-end in CI
- the same engine.js runs in the browser and under the tests

Read more

- docs/BUSINESS_CASE.md — the full one-page business case
- examples/ — two runnable example flows
- README.md — what it is and how to run it offline